HANKANG LEE

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RESEARCH SUMMARY

My research focuses on understanding and designing decision processes in complex operational systems. I examine how local decisions and their interactions shape system-level behavior and outcomes under different operational conditions. I use simulation and data-driven modeling to investigate these relationships and develop decision approaches for coordination and adaptation. I also study how decision processes can expose sensitive information and how privacy requirements shape their design and evaluation alongside performance objectives.

RESEARCH INTEREST

- Decision processes and system-level behavior in complex operational systems
- Multi-agent coordination and adaptive decision-making under uncertainty
- Digital twin and simulation-driven system modeling
- Privacy-preserving learning and trustworthy decision systems
- Human–AI decision-making

EDUCATION

The Pennsylvania State University Aug. 2019 – Aug. 2025
Ph.D. *in Industrial Engineering*

· Advisor: Dr. Hui Yang
· Dissertation: Smart and Autonomous Manufacturing Systems: Design, Modeling and Optimization (https://etda.libraries.psu.edu/catalog/19218hxl5495)

Yonsei University Sep. 2015 – Feb. 2018
M.S. *in Industrial Systems Engineering*

· Advisor: Dr. Chang Ouk Kim
· Thesis: Two-Dimensional Virtual Metrology in Semiconductor Wafer Manufacturing Process

Yonsei University Mar. 2007 – Feb. 2012
B.S. *in Industrial Engineering*

PUBLICATIONS

Journal Articles

1. **Lee, H.**, and Yang, H.* (2025). "Cognitive Digital Twin for Multiobjective Production Scheduling." *Journal of Manufacturing Science and Engineering*, 147(9): 091004. doi: https://doi.org/10.1115/1.4068980
2. **Lee, H.**, Finke, D., and Yang, H.* (2024). "Privacy-Preserving Neural Networks for Smart Manufacturing." *Journal of Computing and Information Science in Engineering*, 24(7): 071002. doi: https://doi.org/10.1115/1.4063728 (**Editor's Pick Award, 2025**)
3. **Lee, H.**, and Yang, H.* (2023). "Digital Twinning and Optimization of Manufacturing Process Flows." *Journal of Manufacturing Science and Engineering*, 145(11): 111008. doi: https://doi.org/10.1115/1.4063234
4. Cheon, S., **Lee, H.**, Kim, C. O.*, and Lee, S. H. (2019). "Convolutional Neural Network for Wafer Surface Defect Classification and the Detection of Unknown Defect Class." *IEEE Transactions on Semiconductor Manufacturing*, 32(2), 163-170. doi: https://doi.org/10.1109/TSM.2019.2902657
5. **Lee, H.**, Jang, J., Ahn, J., and Kim, C. O.* (2018). "Random Forest Method and Simulation-based Effect Analysis for Real-time Target Re-designation in Missile Flight." *Journal of the Korea Society for Simulation*, 27(2), 35-48. doi: https://doi.org/10.9709/JKSS.2018.27.2.035

Peer-reviewed Conference Proceedings

6. **Lee, H.**, and Yang, H.* (2025, June). "Multi-agent Artificial Intelligence to Self-organize Machine Networks for Resilient Manufacturing." In *Proceedings of the ASME 2025 20th International Manufacturing Science and Engineering Conference (MSEC 2025)*, Vol. 2, Article V002T20A006. Greenville, South Carolina, USA. June 23–27, 2025. doi: https://doi.org/10.1115/MSEC2025-155761
7. Shi, X.#, **Lee, H.**, and Yang, H.* (2024, August). "Multi-agent Modeling of Human Traffic Dynamics for Rapid Response to Public Emergency in Spatial Networks." In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 374-380. Bari, Italy. August 28–September 1, 2024. doi: https://doi.org/10.1109/CASE59546.2024.10711522
8. **Lee, H.**, and Yang, H.* (2023, June). "Digital Twin Simulation and Optimization of Manufacturing Process Flows." In *Proceedings of the ASME 2023 18th International Manufacturing Science and Engineering Conference (MSEC 2023)*, Vol. 2, Article V002T07A013. New Brunswick, New Jersey, USA. June 12–16, 2023. doi: https://doi.org/10.1115/MSEC2023-105613

Manuscripts in Preparation

9. **Lee, H.**, and Yang, H.*, "Self-organizing Machine Networks with Multi-agent Reinforcement Learning for Resilient Manufacturing". *Manuscript in preparation*.
10. **Lee, H.**, Finke, D., and Yang, H.*, "Privacy-Preserving Multi-Agent Reinforcement Learning for Adaptive Layout Reconfiguration in Smart Manufacturing". *Manuscript in preparation*.

(*: Corresponding author, #: Undergraduate student researcher)

RESEARCH PRESENTATIONS

1. **Lee, H.**, and Yang, H. "Cognitive Digital Twin for Multi-objective Production Scheduling in Sustainable Manufacturing", *ASME 2025 20th International Manufacturing Science and Engineering Conference (MSEC 2025)*, June 2025, Greenville, SC
2. **Lee, H.**, and Yang, H. "Multi-agent Artificial Intelligence to Self-organize Machine Networks for Resilient Manufacturing", *ASME 2025 20th International Manufacturing Science and Engineering Conference (MSEC 2025)*, June 2025, Greenville, SC
3. **Lee, H.**, Finke, D., and Yang, H. "MxD 20-11-13 Privacy-preserving Analytics for Smart Manufacturing", *MxD Project Final Presentation*, May 2025, Online
4. **Lee, H.**, and Yang, H. "Digital Twinning and Optimization of Manufacturing Process Flows", *2024 INFORMS Annual Meeting*, October 2024, Seattle, WA
5. **Lee, H.**, Finke, D., and Yang, H. "MxD 20-11-13 Privacy-preserving Analytics for Smart Manufacturing", *MxD Project Technical Presentation*, August 2024, Online
6. **Lee, H.**, and Yang, H. "Digital Twinning and Optimization of Manufacturing Process Flows", *2023 INFORMS Annual Meeting*, October 2023, Phoenix, AZ
7. **Lee, H.**, and Yang, H. "Digital Twin Simulation and Optimization of Manufacturing Process Flows", *ASME 2023 18th International Manufacturing Science and Engineering Conference (MSEC 2023)*, June 2023, New Brunswick, NJ
8. **Lee, H.**, and Yang, H. "Digital Twin Simulation and Optimization of Manufacturing Systems", *IISE Annual Conference & Expo 2023*, May 2023, New Orleans, LA
9. **Lee, H.**, Finke, D., and Yang, H. "MxD 20-11-13 Privacy-preserving Analytics for Smart Manufacturing", *MxD Project Final Presentation*, September 2022, Online
10. **Lee, H.**, Finke, D., and Yang, H. "MxD 20-11-13 Privacy-preserving Analytics for Smart Manufacturing", *MxD Project Technical Presentation*, July 2022, Online
11. **Lee, H.**, Finke, D., and Yang, H. "MxD 20-11-13 Privacy-preserving Analytics for Smart Manufacturing", *MxD Project Technical Presentation*, March 2022, Online

12. **Lee, H.**, and Kim, C. O. "2-Dimensional Virtual Metrology by Using Machine Learning Method", *Fall Conference of KSDT (The Korean Society of Semiconductor & Display Technology)*, November 2017, Suwon, Republic of Korea
13. **Lee, H.**, Kim, C. O., Jang, J., Um, M., and Tae, J. "Virtual Metrology in Semiconductor Etching Process", *Spring Conference of KIIE (Korean Institute of Industrial Engineers)*, April 2017, Yeosu, Republic of Korea

RESEARCH EXPERIENCE

The Pennsylvania State University *University Park, PA*
Research Assistant Jan. 2021 – May 2025

At *Complex Systems Monitoring, Modeling & Control Laboratory*
Ph.D. research focused on smart and autonomous manufacturing, including digital twins, decentralized learning, adaptive factory layout reconfiguration, and privacy-preserving AI.

- **Multi-agent Digital Twin**: Developed a simulation-enabled digital twin integrating multi-agent system modeling, metamodel-based optimization, and graph-based operational flow analytics to evaluate and improve decision policies under uncertainty.
- **Cognitive Digital Twin (CDT)**: Designed decentralized multi-agent decision architectures embedding role-specific intelligence to enable adaptive multi-objective coordination in complex operational systems.
- **Self-Organizing Machine Network (SOMN)**: Developed decentralized multi-agent decision policies, implemented via multi-agent reinforcement learning with centralized training and decentralized execution, enabling real-time layout reconfiguration to preserve operational resilience under uncertainty.
- **Privacy-Preserving AI**: Developed Mosaic Neuron Perturbation (MNP), a feature-sensitive, differentially private neural-network training method for mitigating model inversion attacks while maintaining predictive accuracy. Integrated an MNP-derived privacy mechanism into federated multi-agent reinforcement learning for adaptive layout reconfiguration and evaluated privacy protection and operational performance in simulation.

Technical Researcher Sep. 2021 – May 2025

Privacy-preserving Analytics for Smart Manufacturing (MxD Project 20-11-13)

- Led research and technical development as the team's sole graduate assistant and primary technical contributor, developing and evaluating privacy-preserving learning methods for predictive modeling and multi-agent decision-making. Delivered the resulting software, presented interim and final findings to the sponsor, and authored final technical reports.
- Developed the Year 2 research approach and wrote and submitted the successful continuation proposal; led the follow-on project through completion.

Yonsei University *Seoul, Republic of Korea*
Research Assistant Sep. 2015 – Feb. 2018

At *Industrial System Dynamics Laboratory*

- Built a two-dimensional virtual metrology system for semiconductor manufacturing, transforming high-dimensional process sensor data into wafer-level spatial quality maps; applied ANN-based feature selection, simulated annealing, and regression, improving accuracy despite sparse metrology measurements.
- Collaborated on CNN-based image classification research for wafer surface defect inspection, developing k-NN anomaly detection in the learned feature space to flag previously unseen defect classes.
- Developed a real-time algorithm for in-flight missile retargeting, integrating moving-window random-forest trajectory prediction with simulation-based hit-probability estimation; evaluated target selection under changing target conditions and computation and communication delays.

TEACHING EXPERIENCE

The Pennsylvania State University *University Park, PA*

Undergraduate Researcher's Mentor Aug. 2021 – Jun. 2024

***Xiaoru Shi**, Schreyer Honors College*

- Provided step-by-step guidance in research methods, model development, and academic writing for the student's honors thesis and subsequent IEEE CASE paper.
- Thesis Title: Multi-agent Network Modeling for Rapid Response Against Public Emergency in Communities (https://honors.libraries.psu.edu/catalog/8808xxs141)

Undergraduate Researcher's Mentor Jan. 2022 – May 2022

***Leo Hoppe**, **Jason Humphrey**, **Nicholas Wilburn**, **Yiran Zhang**, Senior Design*

- Mentored a four-student senior design team as a PSU CHOT sponsor representative, providing technical feedback on experimental design, IoT data collection, and predictive-maintenance modeling.
- Project Title: Internet of Things for Predictive Maintenance of Medical Equipment (https://sites.psu.edu/lfshowcasesp22/2022/04/28/internet-of-things-for-predictive-maintenance-of-medical-equipment/)

Teaching Assistant Jan. 2020 – May 2020

IE 453 Simulation Modeling for Decision Support

Grader Sep. 2020 – Dec. 2020 / Sep. 2019 – Dec. 2019

IE 505 Linear Programming

Yonsei University *Seoul, Republic of Korea*

Teaching Assistant Sep. 2016 – Dec. 2016 / Sep. 2015 – Dec. 2015

IIE 3104 Production Process Control

AWARDS AND HONORS

Editor's Pick Award, ASME Journal of Computing and Information Science in Engineering (JCISE) *Sep. 2025*
– For the paper, *"Privacy-preserving Neural Networks for Smart Manufacturing"*.

Harry G. Miller Fellowships in Engineering, The Pennsylvania State University *Spring 2025*
– Merit-based graduate travel support for research presentations at ASME MSEC 2025.

Student/Early Career Researcher Travel Award, NSF *Apr. 2023*
– For the 2023 International Advanced Manufacturing Conference.

3rd Place – Problem #2, ASME CIE Hackathon on Automated Testing in Production Planning *Aug. 2021*
– Developed an AI-driven strategy for production scheduling based on test-based engineering data.

BK21 Plus Scholarship, Yonsei University *Spring 2017, Fall 2015*

Yonsei University Teaching Assistant Scholarship, Yonsei University *Fall 2016*

Yonsei University Research Assistant Scholarship, Yonsei University *Spring 2016*

Armed Forces Day Commendation, Republic of Korea Air Force *Nov. 2013*
– Recognized for advancing simulation-based analysis and operational decision support.

Honor Student, Yonsei University *Fall 2011*

Yonsei University Foundation Scholarship, Yonsei University *Fall 2011, Fall 2007*

SERVICE AND LEADERSHIP

Service Activities

Vice president, *American Society for Quality (ASQ), Penn State Student Chapter*, The Pennsylvania State University — 2024 – 2025

Treasurer, *Korean Graduate Student Association*, The Pennsylvania State University, — 2021 – 2022

President, *Mentors Club*, Yonsei University, — 2011

– Led a mentoring program with 100+ mentors supporting 500+ international students in cultural adaptation.

Referee Activities

IEEE Journal of Biomedical and Health Informatics

IISE Transactions

PROFESSIONAL EXPERIENCE

Yonsei University — *Seoul, Republic of Korea*

Department Administrative Coordinator — Sep. 2018 – Jul. 2019

At *Department of Data Science for Semiconductor Manufacturing*

Republic of Korea Air Force (ROKAF) — *Gyeryong, Republic of Korea*

Software Operator (1st Lieutenant) — Jul. 2012 – Jun. 2015

At *Simulation Analysis Branch, Research, Analysis and Evaluation Wing*

- Applied simulation-based decision support models for tactical and strategic planning.
- Modeled a flight time management system to improve scheduling efficiency under mission constraints.

TECHNICAL STRENGTHS

Programming & Tools	Python (PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn), MATLAB/Simulink, C, Git, Linux/macOS, SQL, PowerSim
Machine Learning & AI	Reinforcement Learning (RL, MARL, CTDE), Deep Learning (MLP, CNN), Tree-based Models (XGBoost, Decision Trees), k-NN, SVM
Privacy-preserving ML	Differential Privacy, Federated Learning, Privacy-preserving Training Methods (Mosaic Neuron Perturbation)
Modeling & Analytics	Digital Twin Systems, Agent-based Modeling, Statistical Learning, Simulation- and Data-driven Optimization (Multi-objective, Constraint-based, and Reinforcement Learning-driven; Applications: Scheduling, Resource Allocation), Experimental Design
ML Systems & Workflows	End-to-End ML Pipelines, Cross-platform Integration, Distributed Training